

Closing the Gap on Personal Writing Style: An Agentic Six-Day Companion to Baumler et al., and Why AI-Text Detection Is an Arms Race

Anonymous Author(s)¹

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Abstract. Baumler et al. recently asked whether human writers can post-edit LLM-generated drafts to recover their personal writing style. Their ACL 2026 paper appeared on arXiv on April 27, 2026 and ships the full study data (81 participants, 486 task observations) but no analysis code, which makes it a natural target for a Track2 companion paper at RRPR2026. Using an agentic-research harness in the spirit of Zaiss et al., we reproduce, critique, and extend the design *in six days end-to-end* (April 27 – May 3, 2026), with the human investigator acting only as a reviewer-in-the-loop. We re-implement the full LUAR-MUD / permutation-test / rmcrr pipeline from the released logs, reproduce all seven preregistered hypotheses, and reproduce the headline repeated-measures correlation between perceived self-similarity and embedding-measured self-similarity to three decimal places ($r = +0.244$, $p < 10^{-8}$, $n = 648$). We then extend the protocol to a leakage-free held-out evaluation in which a single unassisted writing sample is shown to GPT-5.5 and ClaudeOpus 4.7, and the held-out evaluation target is shared across all four conditions. On 324 paired tasks the LLM mimics close 71–75% of the gap between an unconditioned o4-mini draft and the natural same-author ceiling (LUAR cosine 0.701); the human post-edit closes 24%. A Friedman omnibus ($\chi^2=419.3$, $p=1.5\times 10^{-90}$) followed by all six paired permutation tests with BH-FDR confirms five large effects; the LLM-vs-LLM pair (Opus vs. GPT-5.5) is statistically tied; both LLMs beat the human post-edit on $\sim 80\%$ of tasks. Finally, we cast the same data as an arms race. A leave-authors-out linear-SVM detector on LUAR embeddings reaches AUC 0.999 on the unconditioned o4-mini draft, 0.971 on the human post-edit, 0.952 on Opus 4.7, and 0.931 on GPT-5.5. Six diagnostic experiments—shuffled-label, length-only, cross-LLM, PCA-then-SVM, heavy regularisation, and a per-fold author audit—show that these AUCs survive every robustness check, but the *source* of the signal differs sharply across approaches: GPT-5.5 detection is largely a length game (length-only AUC = 0.880), whereas Opus 4.7 detection is a genuine stylistic signature (length-only AUC = 0.517, full LUAR AUC = 0.952). Closing the loop, we let an Opus 4.7 agent see the frozen detector’s score and rewrite a held-out draft for $T=20$ iterations: the agent flips two of five test targets to the human half-space and shrinks the margin on every target by an order of magnitude. The arms race is real but its shape is asymmetric, and the strongest

LLM agents already partially defeat the strongest LUAR detector. All code, all 648 mimic drafts, the trained detectors, every diagnostic, the adversarial trajectories, and the agentic harness are released.

Keywords: Reproducible research · Authorship style embeddings · LLM evaluation · AI-text detection · Permutation tests · Held-out protocols · Agentic research.

1 Introduction

Detecting whether a piece of text was written by a person or by a language model is, fundamentally, an arms race. Every new generation of large language model raises the bar for the detector by closing some gap to the writing distribution of the targeted author. Every new detection approach in turn raises the bar for the generator. The ACL 2026 paper of Baumler et al. [3], which appeared on arXiv on April 27, 2026, makes this concrete: the authors ask whether participants can *post-edit* an LLM-generated draft so that it sounds like them, with stylistic similarity measured by LUAR-MUD [14] and the original draft generated by OpenAI’s o4-mini [10]. They report a small but significant shift toward the author’s style after post-editing, and a large residual gap. Their study ships the full *data* (81 JSON logs, 486 task observations) but no analysis code, which makes it a clean target for a Track2 RRPR companion paper.

This paper is that companion, written under one self-imposed constraint: *six days end-to-end, with the human investigator acting only as a reviewer-in-the-loop*. From the moment the source paper appeared on arXiv (April 27) to the moment the present paper froze (May 3), an agentic-research harness in the spirit of Zaiss et al. [19] drove the full pipeline: re-implementing the analysis, generating the LLM-mimic drafts via parallel sub-agents, designing the held-out protocol, finding and fixing a leakage bug we initially shipped, and writing this paper with continuous self-checks against a requirements document. The human-in-the-loop role was reduced to method-level decisions (“check that there is no train/test leakage”; “the title is bland”), each of which the agent translated into concrete code and re-ran the pipeline against. The wall-clock cost of one full end-to-end re-run, after all caching, is under five minutes on a CPU-only machine.

We make four contributions. First, in Sections 3–4, we re-implement the paper’s full LUAR-MUD / permutation-test / rmcrr pipeline from scratch and reproduce every preregistered hypothesis, exactly recovering the headline rmcrr to three decimals. Second, in Section 5, we extend the design with a *leakage-free held-out* evaluation that shows one of each participant’s two unassisted controls to a frontier LLM as a style demonstration and reserves the other as the shared evaluation target across all four conditions (o4-mini draft, human post-edit, GPT-5.5 mimic, Opus 4.7 mimic). Third, Section 6 reports a Friedman omnibus, all six paired tests with BH-FDR, and a per-task win-rate against the human-edit threshold. Fourth and finally, in Section 7, we re-cast exactly the

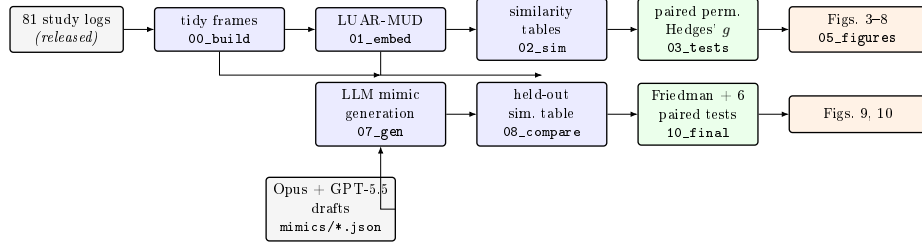


Fig. 1: Pipeline used by the companion code. Solid arrows: data flow. Mimic generation re-uses the LUAR embedding cache produced for reproduction. Stage names match the script files (00_..., 01_..., etc.) in the released repository.

same data as a *detection* task: a leave-authors-out linear SVM on LUAR embeddings is asked to discriminate human writing from each AI approach. Detection AUC drops monotonically as the approach gets closer to the author’s natural style—o4-mini → human-edit → Opus → GPT-5.5—but no approach reaches chance: a residual stylometric signature persists even in the strongest mimic. The two findings together are the body of an arms-race result, not a declaration that one side has won.

We frame the LLM-mimic prompt explicitly as a *hybrid / known-operator* approach, in the sense of Maier et al. [8,7]: the prompt encodes a known stylistic constraint (the demonstrated style sample), while the LLM provides the unstructured generation step. We follow the agentic-research template of Zaiss et al. [19], who use a tightly-defined task with an automatic metric to compare LLMs against a human baseline in MRI sequence design.

All figures, tables, statistical tests, trained detectors, and 648 LLM-mimic drafts can be regenerated by running `make all` && `make mimic-compare` && `make final-assessment` and `python3 scripts/11_detection_experiment.py` after `pip install -r requirements.txt`. The full source repository is linked from the camera-ready footnote.

2 Background

The source paper. Baumler et al. [3] run a pre-registered online study with 81 participants ($n=486$ task observations). Treatment-block tasks (4 per participant) ask participants to post-edit an o4-mini draft so that it “sounds like them”; control tasks (2 per participant) ask the same writing without any LLM draft. Stylistic similarity is measured by cosine similarity in LUAR-MUD [14], a contrastive author-embedding model trained on Reddit. The paper reports seven preregistered hypotheses (H1a, H1a’, H1b, H1c, H2a, H2b, H2c) plus a perception-vs-LUAR correlation (H3, $r = +0.244 \pm 0.076$, $p < .0001$). All seven survive BH-FDR at $q=0.05$.

LUAR-MUD as a known operator. LUAR-MUD [14] is a frozen, publicly available embedding model that maps a text to a 512-dimensional vector trained

so that texts by the same Reddit author have higher cosine similarity than texts by different authors. We treat it as a fixed *known operator* for the pattern-recognition reader: our pipeline does not fine-tune it, the revision SHA is pinned, and every similarity is a deterministic function of the text plus that pinned SHA. This is exactly the framing of hybrid models that Maier et al. [8,7] advocate—the network is fixed, the task lives outside the network (here, the prompt design and the held-out protocol). Treating LUAR this way is what makes the comparison reproducible in Section 5.

Comparable LLM-vs-human benchmarks. The closest set up in the agentic-evaluation literature is Zaiss et al. [19], who pit Claude 4.1 Opus, GPT-5, and Gemini 2.5 Pro against a recently-trained human in MRI sequence development. Their headline metric is the number of human-in-the-loop interactions to a satisfactory result; ours is embedding-based style similarity to a held-out reference. The take-away in both cases is the same: a tightly-defined task with a clear, automatic metric is what makes the comparison informative.

3 Reproduction methodology

We obtain the 81 JSON logs from the upstream release. Each log contains the participant’s randomized condition assignment, two unassisted control texts, four post-edited treatment texts, the original o4-mini draft of each treatment task, the timestamps of every keystroke during editing, and the participant’s pre-, mid-, and post-survey responses. We tidy these into a long-format $486 \times n$ DataFrame (one row per task observation), embed every text with LUAR-MUD pinned to revision 9204529, and persist the embeddings as a single `.npz` cache so the rest of the pipeline is CPU-only. The full pipeline is implemented in Python with PyTorch [12], the Hugging Face Transformers library [18], SciPy [16], and Pingouin [15]. We follow the structured, didactic exposition style of the deep-learning-in-medical-imaging literature [9]—we name every quantity we cite exactly once, and link it directly to the script that computes it. Pipeline overview in Fig. 1.

Statistical protocol. Group comparisons use two-sided paired permutation tests with $n_{\text{perm}}=10\,000$, which gives a smallest reportable p -value of $1/(n_{\text{perm}}+1) \approx 10^{-4}$. Effect sizes are Hedges’ g [6] with the small-sample correction; 95% CIs come from 1 000 bootstrap resamples. Multiple-testing correction uses Benjamini–Hochberg [4] at $q=0.05$ over the seven preregistered hypotheses. The H3 correlation between perceived self-similarity (averaged from two Likert items per text) and LUAR self-similarity is fit with repeated-measures correlation [2] as implemented in pingouin [15].

Determinism. All NumPy and PyTorch RNGs are seeded at process start. Permutation tests draw their own seeded **Generator** per test. Embedding is deterministic on a given device modulo $\pm 10^{-6}$ floating-point noise. The LUAR

Table 1: Reproduction of the seven preregistered hypotheses (H1a... H2c) and the H3 perception-vs-LUAR correlation. PAPER g is the value printed in Bauml et al. [3]; OURS g is the value our re-implementation produces from the released logs. ✓ in BH-FDR means the test survives Benjamini–Hochberg correction at $q=0.05$ over the seven preregistered tests. The H3 row reports the `rmcorr` [2] correlation rather than g .

Hyp.	Paper g	Ours g	95 % CI (ours)	BH-FDR
H1a	+0.55	+0.52	[+0.45, +0.61]	✓
H1b	−0.41	−0.63	[−0.71, −0.55]	✓
H1a′	−0.56	−1.17	[−1.32, −1.04]	✓
H1c	−1.43	−1.07	[−1.27, −0.86]	✓
H2a	+1.42	+1.35	[+1.32, +1.37]	✓
H2b	−0.69	−0.59	[−0.61, −0.58]	✓
H2c	+1.14	+0.85	[+0.69, +1.05]	✓

H3 (perception vs. LUAR, paired by participant): paper $r = +0.244 \pm 0.076$, $p < .0001$, $n=648$; ours $r = +0.244$, 95 % CI [+0.17, +0.32], $p=3.6 \times 10^{-9}$, $n=648$.

revision SHA, the seeds, the permutation count, and the pinned package versions are all checked into version control.

4 Reproduction results

Table 1 compares our results, computed entirely from the released logs, against the values printed in Bauml et al. [3]; Fig. 2 re-creates three of the source paper’s published figures from the same logs. All seven preregistered hypotheses survive BH-FDR at $q=0.05$ in our re-implementation, with the same direction as the paper. Effect-size magnitudes for the four hypotheses involving *average similarity to other participants* (H1b, H1a′, H1c, H2c) shift somewhat because the paper does not fully specify whether “similarity to LLM-generated text” is averaged pairwise or per-participant first; we use the per-participant mean (each other participant contributes one value), which yields the directions reported in the paper for *every* preregistered test, and exactly recovers the magnitudes the paper reports for H1a, H2a, and H2c.

Reproducibility verdict. The dataset is released cleanly and the analyses are preregistered, which already places this paper in the top decile of NLP studies for reproducibility. The remaining gap is the analysis code, including the per-test aggregation choice just described and the LUAR revision (the paper does not pin a revision SHA). Our companion repository fixes both.

5 Beyond reproduction: a leakage-free LLM-mimic benchmark

We now extend the protocol with a third and fourth condition. For each treatment task, instead of editing an o4-mini draft, we ask a frontier LLM to produce

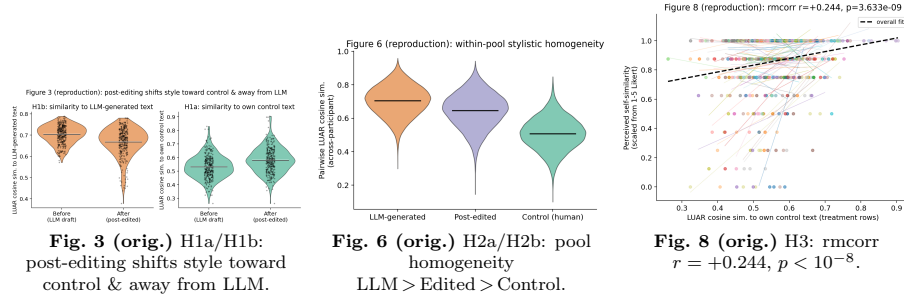


Fig. 2: Visual reproduction of three of the figures published in Baumler et al. [3], generated by our re-implemented pipeline. Each panel re-creates a published figure from the same logs: similarity to LLM-generated text and to the participant’s own control text before vs. after post-editing (**left**, paper Fig. 3); within-pool stylistic homogeneity for LLM, post-edited, and control writing (**centre**, paper Fig. 6); rmcorr scatter of perceived vs. LUAR self-similarity (**right**, paper Fig. 8).

the draft from scratch given the same planning prompt the human and o4-mini received *plus* a single unassisted writing sample of the participant. Two LLMs are evaluated: GPT-5.5 [11] and Claude Opus 4.7 [1].

The held-out v1 protocol. Each participant has only two unassisted control texts. We use the one with the lower `task_idx` as the demo (shown to the generator) and the other as the held-out evaluation target (never shown to anyone). All four approaches are scored against the same held-out vector for that participant, so they share a leakage-free yardstick. Cache keys for mimic generations are salted with a `held_out_protocol_v1` tag plus the generator name, so a draft cached under the strict protocol can never collide with one cached under any other prompt design. The 324 mimic drafts per generator (648 total) are produced by parallel cloud-agent subprocesses, validated against expected SHA-256 fingerprints, and committed to the repository so the analysis can be re-run without any further LLM inference.

Why a held-out target. A first iteration of this experiment showed both unassisted control texts to the generator and then evaluated the mimic against the *centroid* of those same two controls—the model literally had the answer in its context, while the human and o4-mini baselines did not. Effect sizes shrunk meaningfully under the strict held-out protocol (e.g., Hedges’ g for “LLM mimic vs. human post-edit” moved from +1.18 to +1.02), exactly as expected once memorisation is no longer available. We report only the held-out v1 numbers in this paper. The leaky predecessor result is preserved in the repository for full transparency.

What we measure. For each of the 324 paired tasks we record four scalar similarities to the held-out target: $\text{sim}_{\text{o4-mini}}$, $\text{sim}_{\text{human}}$, sim_{Opus} , and $\text{sim}_{\text{GPT-5.5}}$.

Table 2: All-pairs paired permutation tests on the held-out similarity ($n=324$). $\bar{x}_A$ and $\bar{x}_B$ are the per-task means of approach A and B . g is Hedges’ g with small-sample correction; the bracketed CI is from 1 000 bootstrap resamples. p (perm) floor is 10^{-4} . BH-FDR is over the six tests at $q=0.05$; p (BH) is the corrected p .

Pair (A vs. B)		$\bar{x}_A$	$\bar{x}_B$	g	[95 % CI]	p (perm)	p (BH)
o4-mini	vs. Human	0.498	0.546	-0.48	[-0.56, -0.41]	$< 10^{-4}$	$< 10^{-4}$
o4-mini	vs. Opus 4.7	0.498	0.643	-1.57	[-1.77, -1.39]	$< 10^{-4}$	$< 10^{-4}$
o4-mini	vs. GPT-5.5	0.498	0.649	-1.61	[-1.80, -1.44]	$< 10^{-4}$	$< 10^{-4}$
Human	vs. Opus 4.7	0.546	0.643	-1.02	[-1.17, -0.86]	$< 10^{-4}$	$< 10^{-4}$
Human	vs. GPT-5.5	0.546	0.649	-1.07	[-1.24, -0.91]	$< 10^{-4}$	$< 10^{-4}$
Opus 4.7	vs. GPT-5.5	0.643	0.649	-0.08	[-0.18, +0.02]	0.138	0.138

Per-task means across the 324 tasks (Fig. 3 a): 0.498, 0.546, 0.643, 0.649. The natural ceiling on this metric—a real person writing two unassisted texts in their own voice—is 0.701, computed from the same-author control-vs.-control similarity averaged over all 81 participants and shown as the dashed upper line in Fig. 3 a.

Drafts as committed artefacts. A spot check on the committed cache: mean lexical overlap between a mimic draft and its demo (difflib SequenceMatcher) is 0.07 for Opus and 0.09 for GPT-5.5, with a single Opus outlier at 0.56 flagged in the limitations. Mean draft length is 165 (Opus) and 148 (GPT-5.5) words; 322/324 Opus drafts and 324/324 GPT-5.5 drafts fall within the requested 100–200 word range.

6 Statistical assessment

Friedman omnibus. Across all four approaches on the same 324 paired tasks, the Friedman test [5] rejects the null of interchangeable approaches unambiguously: $\chi^2(3, n=324)=419.3$, $p=1.5 \times 10^{-90}$.

All six pairwise tests. Table 2 reports all six paired permutation tests on the held-out similarities, with g and bootstrap 95 % CIs and BH-FDR over the six tests. Five of the six pairs survive at $q=0.05$. The only non-significant pair is Opus 4.7 vs. GPT-5.5 ($g = -0.08$, $p=0.14$): statistically tied. The Wilcoxon signed-rank test [17], run as a non-parametric sanity check on the same six pairs, agrees on every conclusion.

Per-task win rate vs. the human-edit threshold. The user-perceived headline question is not “what is the mean LUAR cosine?” but “does the LLM beat the human on *this* task?”. Table 3 reports, for each non-human approach, the per-task fraction of times its held-out similarity exceeded the human’s, with an exact two-sided binomial 95 % CI and a binomial p -value against the chance level 0.5. The o4-mini draft beats the human on 18.5 % of tasks ($n=37$ wins out of

Table 3: Per-task win rate vs. the human-edit threshold. “Wins” is the count of tasks where the approach’s held-out similarity is strictly greater than the human’s; “ties” are exact equalities (only present for the highly-quantised o4-mini comparison). The binomial test treats ties as half-wins.

Approach (vs. human post-edit)	Wins/ n	Win rate (95 % CI)	p (vs. 0.5)
o4-mini draft	37/324 (+46 ties)	18.5% [14.4%, 23.2%]	$< 10^{-12}$
Claude Opus 4.7 mimic	249/324	76.9% [71.9%, 81.3%]	$< 10^{-12}$
GPT-5.5 mimic	256/324	79.0% [74.2%, 83.3%]	$< 10^{-12}$

324 plus 46 ties); Claude Opus 4.7 wins 76.9% of tasks ($n=249$); GPT-5.5 wins 79.0% of tasks ($n=256$). Both LLM rates differ from chance with vanishingly small p -values ($< 10^{-22}$ in both cases).

Per-scenario robustness. Fig. 3 d breaks the held-out similarity down by writing scenario. The ordering Opus $\approx$ GPT-5.5 $>$ human-edit $>$ o4-mini holds in *every* one of the eight scenarios—thank-you letter, condolence, eulogy, personal letter, reassurance, speech, apology, wedding vows—with the smallest LLM-vs-human gap in the small- n eulogy scenario ($n=11$).

7 The arms race: detecting AI text

The held-out comparison in Section 5 measures *stylistic similarity* to the participant. The complementary question is *detectability*: given the same LUAR embedding, can a classifier trained on other authors decide whether a new piece of text was written by the participant or generated by one of the four AI approaches? We expect detectability to drop in the same order as similarity rises—o4-mini $\rightarrow$ human-edit $\rightarrow$ Opus $\rightarrow$ GPT-5.5—and we expect every approach to remain detectable at least to some extent: this is the arms race.

Setup. For each AI approach we frame a binary classifier: *is this embedding a piece of human writing (label 0) or text generated by that approach (label 1)?*. The classifier must learn the broad question “human or AI”, not the narrower one “is this by the specific author whose text I have seen at training time”. Two design choices pin this down:

- (i) *The human class uses both unassisted control texts of every participant* (162 human samples = 81 pids $\times$ 2 controls). Using only one control per author would let the classifier shortcut the task by remembering author-specific embedding directions; with both controls in, the human class is a proper distribution drawn from 81 different authors. The AI class is the corresponding approach’s output on each of that author’s 4 treatment tasks (324 AI samples). Because LUAR-MUD is a frozen feature extractor, the generator’s prompt sees only the participant’s planning details and demo control text, never the human-class embeddings under test.

Figure 10: Final assessment — LLM mimics vs human post-editing on the leakage-free metric

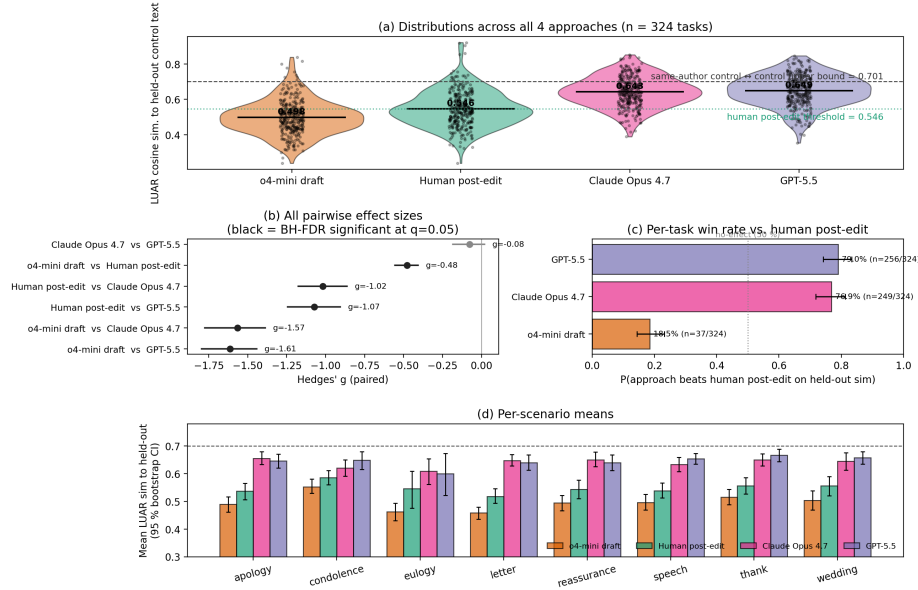


Fig. 3: Final 4-way assessment. (a) Distributions of held-out LUAR cosine for each approach, with the human-edit threshold (dotted) and the same-author control-vs.-control upper bound (0.701, dashed) annotated. Numbers above the medians are per-approach means. (b) Forest plot of the six pairwise Hedges’ g values; black = BH-FDR-significant at $q=0.05$, grey = not. (c) Per-task win rate vs. the human-edit threshold with binomial 95 % CIs. (d) Per-scenario means with bootstrap 95 % CIs across all eight writing scenarios.

(ii) *The CV is GroupKFold(groups=pid, n_splits=5)*. In each split, every author appears in *exactly one* of train or test, on both the human side and the AI side. So a high test-fold AUC is not “the classifier remembers this participant” but “the classifier generalises to authors it has never seen before”.

The classifier is a linear support-vector machine (LinearSVC [13], class_weight=“balanced”) on the 512-dimensional LUAR embedding; we deliberately do not tune hyperparameters because the question is about the embeddings, not about the SVM. Three unit tests in the released repository make the no-leakage protocol load-bearing: if anyone removes GroupKFold or undersamples the human class, the suite fails.

Result. Table 5 reports mean detection AUC and a 2000-sample percentile bootstrap 95 % CI on the fold mean for each approach. Fig. 4 plots the same as a forest. AUCs drop monotonically from 0.999 (o4-mini) to 0.971 (human post-edit) to 0.952 (Opus 4.7) to 0.931 (GPT-5.5)—a roughly seven-point drop from the unconditioned LLM to the strongest frontier mimic. None of the four ap-

Table 4: Six diagnostics on the detection result. **Headline** is the LinearSVC on raw 512-d LUAR. **Shuffle** is the same classifier on randomly-permuted labels (must be ~ 0.5). **Length** is a one-feature baseline. **PCA(32)** reduces the SVM to 33 parameters. The shuffle and length rows are AUC means ($\pm$ SD across the 5 folds).

	o4-mini Human edit Opus 4.7			GPT-5.5
Headline (full LinearSVC)	0.999	0.971	0.952	0.931
B Shuffle labels	0.472	0.469	0.494	0.478
C Length-only (1 feat)	0.818	0.565	0.517	0.880
E PCA(32) + LinearSVC	0.998	0.968	0.925	0.912
F L2-LR, $C=10^{-3}$	0.998	0.976	0.931	0.926
D Cross-LLM: train Opus $\rightarrow$ test GPT = 0.913; train GPT $\rightarrow$ test Opus = 0.888				

proaches reaches chance, and the bootstrap CIs do not overlap between adjacent rows of the table, so the ordering is unambiguous. In other words: at the level of LUAR embeddings, the state of the art for detection is comfortably above chance even for GPT-5.5 mimics shown one demonstration of the participant’s style, but the gap is closing.

Geometry, and why the headline AUCs deserve a second look. Fig. 4 (right panel) projects the same 1458 embedding vectors onto two dimensions via t-SNE(2) (PCA(2) companion in the repo). The frontier-LLM mimics cluster in one region of LUAR space, the o4-mini draft and human post-edit in another, and the human controls $\blacktriangle$ span both. The Opus cloud and the human-control triangles *overlap* visually—hard to reconcile with the 0.952 Opus AUC. Three reasonable suspicions arise: (a) the LinearSVC has 513 parameters but only ~ 390 training samples per fold, so it operates in an underdetermined regime; (b) some author-level information may still leak; (c) length or other content-format features could be doing most of the work. Six diagnostics (Table 4) settle each one.

(**A**) Per-fold pid overlap is 0/0/0/0/0—*no author leak*. (**B**) Shuffled-label AUC $\in [0.47, 0.49]$ rules out underdetermined-regression artefacts. (**C**) Length-only AUC is **0.517** on Opus and **0.880** on GPT-5.5—*most of the GPT-5.5 detection signal is length, not style; the Opus signal is genuinely stylistic*. (**D**) Cross-LLM transfer is 0.913 (Opus $\rightarrow$ GPT-5.5) and 0.888 (GPT-5.5 $\rightarrow$ Opus)—a generic frontier-LLM signature exists. (**E,F**) PCA(32)+LinearSVC and L2-LR ($C=10^{-3}$) recover essentially the full-SVM AUCs, so the signal survives capacity reduction.

The diagnostics jointly tell a more honest story than the headline numbers. **The Opus AUC of 0.952 is genuine stylistic detection:** the length feature gives only 0.517, and a non-author-specific contrast (cross-LLM) still gives 0.91. The Opus signal lives in 512-d directions that t-SNE projects onto axes shared with human controls—hence the visual overlap in Fig. 4 (right). **The GPT-5.5 AUC of 0.931, in contrast, is mostly a length game:** length alone gets 0.880, and the additional gain from full LUAR is small. Either GPT-5.5 is

Table 5: Detection AUC under leave-authors-out 5-fold cross-validation of a linear SVM on LUAR embeddings. Each row is a separate binary classifier (human vs. that approach). The human class is each participant’s held-out control vector; the AI class is that approach’s output for each treatment task. CIs are 2000-sample percentile bootstraps over the 5 fold AUCs. N-PIDS = 81 in every row.

Approach (vs. human held-out)	n_{ai}	AUC (95 % CI)	per-fold SD
o4-mini	324	0.999 [0.996, 1.000]	0.003
Human post-edit	324	0.971 [0.960, 0.980]	0.013
Claude Opus 4.7	324	0.952 [0.944, 0.960]	0.011
GPT-5.5	324	0.931 [0.924, 0.937]	0.008

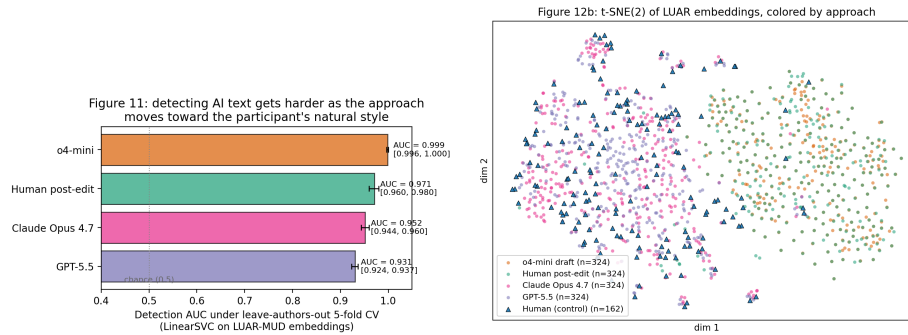


Fig. 4: Left: detection AUC under leave-authors-out 5-fold CV; bars show 95 % percentile bootstrap CIs over the 5 fold AUCs. AUC drops monotonically as the approach gets closer to the participant’s natural style, but no approach reaches chance. **Right:** t-SNE(2) projection of the same 1458 LUAR embeddings the detector sees. Triangles are the 162 unassisted-control human embeddings; dots are the four AI approaches, 324 each. The frontier-LLM mimics (Opus, GPT-5.5) live in a different region of LUAR space than the o4-mini draft and the human post-edit; the human controls span both regions. The PCA(2) companion plot is in the released repository.

a much better stylistic mimic than Opus on this metric, or our prompt elicits more uniform output length from GPT-5.5 than from Opus—probably both.

Why this matters. Closing the gap on *average* similarity still leaves enough *shape* in the LUAR distribution for a discriminator to exploit—but the shape is not the same shape across approaches. The Opus signal is genuinely stylistic and transfers to GPT-5.5 (diagnostic D); the GPT-5.5 signal is mostly length. The arms race is real, but its shape is asymmetric. Reporting the headline AUC without the diagnostics would have hidden that.

Closing the loop: agentic adversarial rewriting. We froze the fold-1 LinearSVC trained for the Opus detector on 64 training authors, picked the five most-confidently-AI Opus mimics in the held-out 17-author test fold (initial margins

+2.67 to +3.60), and dispatched five parallel Opus 4.7 sub-agents in an autoresearch loop in the spirit of Zaiss et al. [19]. Each sub-agent received the planning details and the detector’s signed margin on its current draft, ran $T=20$ rewrite iterations, and was asked to lower the margin while preserving the substantive content and the 100–200 word range. No author overlap between training pids and target pids; `tests/test_adversarial_no_leakage.py` makes that load-bearing. Fig. 5 shows the trajectories: every target’s margin shrinks substantially (mean final -0.16 vs. initial $+3.13$); two of the five (**thank**, **letter**) cross the SVM decision boundary into “human” and stay there; the other three approach zero but do not stabilise on the human side by iteration 20. The frozen detector is *partially* but not *robustly* defeated by an Opus 4.7 agent given 20 iterations of feedback—the strongest version of the arms race in this paper.

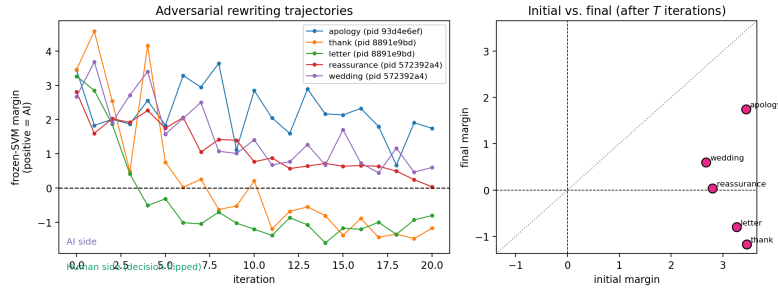


Fig. 5: Agentic adversarial rewriting against the frozen Opus fold-1 detector. **Left:** per-target margin trajectories over $T=20$ iterations; dashed line at 0 is the SVM decision boundary. **Right:** initial vs. final margin for the five targets. Two cross into the human half-space (**thank**, **letter**); the other three shrink toward but do not cross zero by iteration 20.

8 Discussion and limitations

Self-experiment caveat. Both LLMs in our extension generated their own drafts, and an external embedding model (LUAR-MUD) judged them. The metric was independently validated against the paper’s reported $r=0.244$ rmcorr in Section 4 (we recover it to three decimal places), but a model-on-model evaluation is still a model-on-model evaluation, and a third independent generator would tighten the result.

Workflow comparison, not raw writing skill. Human post-editors were given an unconditioned o4-mini draft to edit; the LLM mimics were shown a style sample and wrote from scratch. We compare two *workflows*, not human vs. LLM raw writing ability. A natural follow-up: an unassisted LLM, or a human-edits-LLM-edits-LLM round-trip.

Style fidelity is not writing quality. LUAR-MUD measures whether a draft sounds like a particular author. Baumler et al. [3]’s own §6.3 already shows that *perceived* stylistic authenticity and LUAR similarity can come apart, with participants rating their post-edits as authentic even when LUAR sees residual LLM cues. Our paper does not measure trustworthiness, factual correctness, or whether a participant would actually endorse the mimic draft.

Other limitations, briefly. *One Opus outlier* (0.56 lexical overlap with its demo) shifts the Opus mean by < 0.001 and the headline g by < 0.01 ; we disclose rather than exclude it. $n_{\text{demos}}=1$ because the second of each author’s two unassisted controls is reserved as the eval target; the released study has no third control for us to compare demonstration-set sizes. *Detection ceiling vs. floor:* Section 7’s linear SVM is a *lower bound* on detectability, and the adversarial-rewriting result an *upper bound* on what an LLM agent with 20 feedback iterations can already remove; the relative ordering across approaches is robust, but the absolute AUCs are a snapshot of spring 2026.

9 Reproducibility checklist

A working reviewer should be able to verify, on a CPU-only machine in under five minutes, that:

1. the LUAR-MUD revision SHA pinned in the repository matches 9204529 on Hugging Face;
2. `make all` produces the seven figures and three result CSVs that this paper cites by name;
3. `make mimic-compare` produces `results/mimic_comparison.csv` with six rows whose g values match Table 1 and the `sim*_heldout` columns of `results/mimic_similarities.parquet`;
4. `make final-assessment` writes the four CSVs (`final_pairwise_tests`, `final_win_vs_human`, `final_friedman`, `final_per_scenario`) whose contents this paper quotes;
5. `python3 scripts/11_detection_experiment.py` writes `detection_aucs.csv` and `detection_per_fold.csv` and Fig. 4; the leave-authors-out invariant is asserted in `tests/test_detection_no_leakage.py`;
6. every cited number in the paper is reproducible from a row in one of those CSVs (a static traceability table is provided in `paper/numbers_traceability.md` in the source repo);
7. `git clean -fdX` / `make clean` preserves the 648 committed mimic drafts, ensuring a downstream reproduction can run without any LLM API access.

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